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A new explanation for the WTP/WTA disparity

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Abstract

We propose a new explanation for the WTP/WTA disparity in experiments and surveys. Uncertainty, irreversibility and limited learning opportunities can generate commitment costs, driving a wedge between WTP and WTA. We present experimental evidence that supports our hypothesis.

Keywords: Willingness-to-pay; Willingness-to-accept; Commitment cost; Lab experiment; Contingent valuation survey

JEL classification: D60; D83; C90

1. Introduction

Empirical disparities between willingness-to-pay (WTP) and willingness-to-accept (WTA) measures observed in stated preference surveys and laboratory experiments have become well known. (See Horowitz and McConnell, 2000a for a nice review). Hicksian welfare theory cannot explain this divergence since the implied income effects are implausibly high (Horowitz and McConnell, 2000b). One popular explanation concerns the notion of reference-dependent preferences (Kahneman and Tversky, 1979 and Tversky and Kahneman, 1991), also known as loss aversion or endowment effects (Thaler, 1980; Morrison, 1997, 1998). This theory suggests that a consumer's preference is directly a function of her initial endowment; thus changes in endowments alter the marginal rates of substitution between goods. A second explanation is due to Hanemann (1991): building on Randall and Stoll (1980), he argues that the lack of substitutes can generate large divergences between the compensating variation (CV) and equivalent variation (EV), and thus WTP and WTA. Kolstad and Guzman (1999) further argued that the divergence can be caused by the auction design used in experiments.

In this paper, we present an explanation based on the effect that timing of an action can have on

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WTP and WTA. We argue that even if $CV = EV$, WTP can still be substantially smaller than WTA. Specifically, if a consumer is unsure of the value of the good (measured by CV or EV), and there are non-trivial transaction costs associated with reversing her purchasing or selling decision, she may prefer to delay the decision in order to obtain additional information about the good's value. If she must announce a WTP (WTA) value today, thereby giving up the option to gather more information, she will require compensation for the lost learning opportunity. The result will be a lower WTP (higher WTA) than the true expected value of the good (i.e. CV or EV).

A large literature has developed explicitly incorporating the consequences of uncertainty, learning and irreversibility into investment decisions (Arrow and Fisher, 1974; Dixit and Pindyck, 1994). The fundamental result is that these factors often lead a decision maker to delay investment, and the value of the investment depends on its timing (or the information available). Though there has been little application of these concepts to the consumer demand arena, the underlying intuition can be readily extended to the situation where the consumer has to act now but can change her bid or ask price in response to these factors. Then a natural conclusion is that the time at which (or the amount of information with which) WTP and WTA are formed may affect their magnitude.

These observations provide an additional explanation for the WTP/WTA divergence in experiments and surveys. Specifically, these 'anomalies' can arise when experiments and surveys restrict the time and opportunity a subject has to gather relevant information. Then the WTP/WTA disparity is the logical consequence of irreversibility (or adjustment costs), uncertainty and timing of the transaction, even when the subjects are rational and have neoclassical preferences.

2. A real options explanation for WTP/WTA divergences

In this section, we briefly discuss the conditions under which the WTP/WTA divergence will arise, and how these conditions can be generated in experiments and surveys. Our discussion will concentrate on the intuition involved (the formal model can be found in Zhao and Kling, 2000).

Suppose an agent is considering purchasing a product that cannot be returned. She does not know the product's value to her but knows its distribution. The expected value is her CV, and in a static framework, is also the maximum price she is willing to pay, i.e. her WTP. But suppose the consumer can learn more about the good's value later, say through consulting with her friends. Further after she does so, the product (or a close substitute) will still be available. Quite intuitively, if the price exactly equals her expected value, she is not willing to buy the product *now*. Rather, she will wait for her friends' opinion, so that she can make a more informed decision and avoid the losses she would incur if the value to her turns out to be below the price. Of course, waiting incurs costs: in this case, the consumer has to delay her consumption till a later date. She needs to balance her waiting cost against the expected benefit of more information. Her incentive to wait decreases as the price is lower, since the probability of making a purchase she regrets decreases. In the limit, if the price is zero, she will buy the product today since the purchase simply cannot be a bad deal. Consequently, there exists a unique price at which the consumer is indifferent between buying now and waiting, and this price is the consumer's maximum WTP. This WTP is generally lower than her CV, and the difference measures the value of her option of obtaining more information, which we term the commitment cost and denote as CC_p. That is, $WTP = CV - CC_p$.

By a similar argument, if an agent is considering selling a product whose value she is unsure of, and if she can obtain more information about the value later, she will demand a higher compensation than her EV to sell the good *now*. That is, $WTA = EV + CC_A$, where CC_A is the commitment cost associated with the selling decision.

The two commitment costs thus generate divergences between WTP and WTA over and above that between CV and EV. Thus even if $CV = EV$, WTP and WTA can still diverge by $CC_P + CC_A$. Either one of the commitment costs can be significant relative to the true CV and EV. For example, suppose the good's value is distributed over $[0, G]$ with mean $G/2$, and there is no income effect. Then $CV = EV = G/2$. Further, suppose the consumer is perfectly patient: she is not planning to use the product now and can wait for her friends' suggestions without any cost. Then her $WTP = 0$: for any price higher than zero, there is always a probability that the purchase is not worthwhile so there is always positive expected gain from learning. Similarly, her $WTA = G$ if she does not immediately need the proceeds of selling the good. Then the WTA/WTP ratio is infinite and the absolute difference is twice the value of the good!

According to this explanation, the magnitude of the WTP/WTA divergence depends on the size of the two commitment costs. Thus we can make predictions about the divergence by studying how commitment costs are affected by the characteristics of an experiment or survey. The testable predictions from our explanation can be summarized as:

2.1. Hypothesis 1

The observed divergence between WTP and WTA decreases as the subjects (1) *are less uncertain about the good's value*, (2) *expect that less information can be gathered in the future about the good*, (3) *are more impatient in consuming the good or the proceeds of selling the good*, (4) *expect that reversing the transaction becomes easier, and* (5) *have more freedom in choosing when to make the decision*.

The hypothesis also lists the conditions needed for the commitment costs to arise. Now we consider how experiments and surveys might generate these conditions. In experiments, the traded good might be a chocolate bar or a coffee mug. Sometimes the chocolate bars are fancy, specialty bars and the mugs have university logos or other distinctive features. Thus, typical subjects may not be certain of their valuation of these items. Even if the traded good is a common item, a subject may be uncertain about its price in the stores outside the experiment and the prices and values of its substitutes and complements.[1] Further, the degree of uncertainty needed to generate a significant divergence between WTA and WTP is not particularly large. For example, suppose the traded good is a mug and a subject believes that its mean value is about \$6.00, but the true value can be either \$4.00 or \$8.00 with equal probability. If she does not have to use the mug now and can discover its true value later, the WTP and WTA are \$4.00 and \$8.00, respectively, generating a 100% difference between the two.

Next, our explanation assumes that the agent can delay her decision to gather information and still have the same trading opportunities. In many lab experiments and surveys, the subjects are given

[1] It should be noted that typically subjects are 'broadly familiar' with these traded goods in experiments, at least with the category of these goods. However, 'broad familiarity' with a good is far different from knowing precisely (or even approximately) the present value of all future consumer surpluses associated with the good, and the price information in all local stores about both the good and its substitutes or complements. Uncertainty required for our explanation is about the latter condition.

choices only during the experiment or survey. It may thus appear that they do not have the ability to delay the trading decision. However, if the good used in the experiment is available in regular markets, subjects clearly have the ability to delay in the purchase decision: instead of buying the good during the experiment, she can do so in a regular market after the experiment. Even if the good is not available outside the experiment, there may be *substitutes* with similar functions in regular markets that the subjects can buy after the experiment. Also, in this situation, reversing the purchase made in an experiment or survey is costly: a subject's decision is essentially irreversible within the experiment and survey sessions. She must find ways to reverse the trade outside these sessions. Resale markets will generally be quite poor for goods that are typically purchased in experiments.

On the other hand, there may be situations in which it is difficult to purchase the good outside of the experiment setting. In that case, the commitment cost will be more significant in the WTA formation. This might be the case for unique environmental goods or consumer health. For example, a subject considering selling a national park may perceive it to be impossible to re-purchase the park, generating an enormous commitment cost. Note that the existence of one commitment cost is enough to generate the WTP/WTA divergence. In fact, the very high divergence between WTP and WTA observed for some environmental goods, combined with unintuitively high WTA's, is consistent with a large commitment cost associated with WTA.

The goods used in experiments and surveys are typically not urgently needed by the subjects. Compared with an actual market where consumers are actively seeking to purchase a good, subjects in an experiment or survey generally do not go to the laboratory with the intention of buying or selling a chocolate bar or coffee mug. Thus, it is likely that subjects would be very willing to wait to make trades if allowed.

In sum, experiments and surveys have two distinctive features compared with real markets: limited time and limited learning. While in a real market a consumer may take time to gather information about the good before making her decision, a respondent in an experiment or survey must report or act upon a *WTP* or *WTA* value during the experiment or survey. By *forcing* the respondents to make decisions before they *voluntarily* stop information gathering, experiments and surveys potentially increase the commitment costs, and thus the divergence between *WTP* and *WTA*. If this effect is significant, then the *WTP*/*WTA* divergence are mainly artifacts of the experiment and survey design, rather than indicating the failure of expected utility or Hicksian welfare theory.

3. Pre-existing evidence

The predictions from Hypothesis 1 have not been directly tested in the literature. But there is some evidence from published experimental studies that lends support to some of them. In analyzing the following pre-existing evidence, it is important to keep in mind that more than one explanation for the WTP/WTA divergence may be present.

3.1. Level of uncertainty

In an insightful paper, Kahneman et al. (1990) (KKT hereafter) conducted a series of experiments to investigate the persistence of endowment effects in repeated experiments. First, several experiments (called markets) are conducted for induced-value tokens: a token can be cashed at a *predetermined* price at the end of the experiment, thus no uncertainty in the token's value exists. Next, a series of constructed markets are conducted for university coffee mugs, boxes of ballpoint pens, and folding binoculars, all available in university bookstores. KKT found no WTP/WTA divergence in token markets and persistent divergence in the other markets.

The token market results are consistent with our model predictions: since there is no uncertainty about the token's value, both commitment costs, CC_P and CC_A, are zero and there will be no WTP/WTA divergence. Further, we would predict that if the token is not for a *predetermined* cash amount, but for something with uncertain values, such as a lottery of cash payments, then the WTP/WTA divergence will arise. We are unaware of experiments that directly test this hypothesis, but in an early experiment reported in Knetsch and Sinden (1984), a lottery ticket is directly traded and WTP/WTA divergences do arise, lending some support for our prediction.

List (2000) conducted a series of experiments where subjects trade sports cards, and found that the divergence decreases for subjects with more market (or trading) experience. To the extent that market experience reduces a subject's uncertainty about the value of the traded card, his findings are consistent with the commitment cost explanation. He also found that the divergence for sports card dealers is significantly smaller than that for non-dealers. Since dealers typically have more opportunities than non-dealers to reverse their transactions (to resell their purchased cards or to purchase the sold cards at later dates), the commitment costs for dealers should be smaller, leading to a smaller divergence.

3.2. Learning within an experiment

KKT, in each experiment, repeated the constructed market four times to allow subjects to learn about the trading institution and found that the WTP/WTA divergence persisted. In the commitment cost story, however, 'learning' is not only about the trading institutions, but more importantly about the value of the traded good, its substitutes and complements, and, generally, anything related to making the trading decision. Since there is no reason to expect that subjects will learn adequately about these values through repeated experiments, our model predicts a persistent divergence, which is consistent with their results.

For experiments that do provide adequate learning opportunities about the value of the traded good, we would expect the WTP/WTA divergence to decline over repeated trials. An example of such an experiment is found in Coursey et al. (1987) (CHS hereafter). CHS conducted repeated experiments where the traded *good* is not having to taste (specifically, to hold in the mouth for 20 s) a cup of a very bitter chemical, sucrose octa-acetate (SOA).[2] Before the formal repeated trials, subjects tasted 'a few sample drops' of SOA, forming the priors about the good's value. After each trial, the prevailing market price (formed from a fifth price Vickrey auction) is posted and *subjects (in particular any single winner) are allowed to demand another trial*, subject to a limit of 10 trials in total. CHS concluded that the WTP/WTA divergence essentially disappears in this experimental setting.

The experiment is quite unique in terms of its learning environment. First, learning about the good's value can occur only within the experiment, since the good (or a substitute) is not available otherwise. Here, learning is achieved through observing posted prices, which represent how other subjects in the experiment value the good. These subjects are the only people who have tasted a

[2] Thus, buying the good means paying money to avoid tasting SOA while selling means getting paid for tasting SOA.

sample and are thus knowledgeable about the good. Their valuation provides the only possible additional information about the good's value.[3] Second, in many cases (when the experiment stops before the tenth trial), stopping learning and making a trade is a voluntary choice of the subjects. In these cases, subjects are not *forced* to make trading decisions within a time limit.

Based on the commitment cost story, subjects would voluntarily stop learning when commitment costs are small. Our theory thus predicts that the WTP/WTA divergence would become small or disappear if voluntary stopping occurs before the tenth trial. However, if the tenth trial is reached, our model would predict a larger divergence since learning has not *voluntarily* stopped, and commitment costs may still exist. Both predictions are confirmed by CHS's results. Average WTA and average WTP are strikingly similar for experiments where subjects voluntarily cease trading before the tenth trial. For the only two experiments where WTA and WTP, respectively, are elicited in the tenth trial, the WTP/WTA divergence is significant.[4]

3.3. Degree of patience

In most experiments the goods traded are not likely to be what a subject had in mind to purchase that day. For example, it seems unlikely that many subjects will enter an experiment only to discover that the coffee mug they had planned to purchase later in the day is being traded in the laboratory. Subjects without a keen desire to purchase a coffee mug *now* will exhibit a high degree of patience and the associated commitment costs may be high, leading to big WTP/WTA divergence.

Knetsch and Sinden (1984) briefly report (in their footnote 3) an experiment where the subjects are less patient. The traded good was a lunch and the experiment was 'carried out as respondents entered an office cafeteria.' Since the subjects are just in need of the traded good, delaying is costly and their patience level is low. Thus our explanation predicts low commitment costs and a small WTP/WTA difference. This prediction is confirmed by the experimental result: Knetsch and Sinden (1984) found that the WTP/WTA difference was not statistically significant.[5]

3.4. Additional reference points

Bateman et al. (1997) report the results of a carefully designed study to investigate additional predictions of reference-dependent preferences, above and beyond the divergence between WTP and WTA. They investigate four different reference points from which subjects initiate trades and show

[3] It might be argued that since subjects have tasted the good once, no additional learning is necessary or possible. However, the 'good' to be consumed is to hold the SOA in the mouth for 20 s which may be quite a different experience from simply having a quick taste. Further, even if a subject knows how bitter SOA is, she may still be unsure about its *value*.

[4] CHS did not report these two findings directly. We base our observation on their Fig. 1, where WTA2–WTA4 and WTP1–WTP3 are elicited before the tenth trial and WTA1 and WTP4 are elicited in the tenth trial. Some researchers have argued that the use of a Vickrey auction with announced prices after each trial leads to WTP/WTA convergence since the announced prices serve as anchors for subjects. Notwithstanding the anchoring effect, the fact that the WTP/WTA divergence occurs only for the two experiments where trading happens in the tenth trial still lends some support for the commitment cost hypothesis.

[5] Knetsch and Sinden seemed to recognize this 'patience' effect. They guessed that a possible reason for the convergence is 'the fact that the respondents were already committed to spending money for a lunch at the time of their participation in the experiment.'

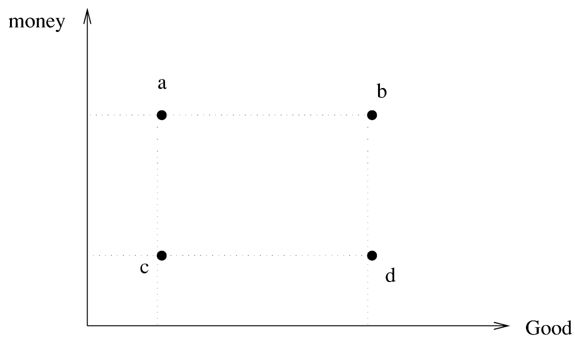

Fig. 1. Reference points in Bateman et al's experiments.

that divergences arise for some, but not all of the four cases. Again, these results are consistent with the predictions of the commitment cost explanation.

Bateman et al. (1997) define four measures, each uniquely identified with a reference point in Fig. 1. At points a and d, the conventional WTP and WTA measures, denoted as WTP_a and WTA_d, are elicited for the traded good (four cans of Coke and later 10 chocolate bars). At point b, 'equivalent loss' EL_b is elicited which measures how much a subject is willing to give up to avoid losing the good. At point c, 'equivalent gain' EG_c is used to measure how much a subject demands to be compensated for not getting the good.

Our explanation predicts that $WTP_a = EL_b$ since the fundamental decision a subject makes in both scenarios is to give up money for the good. Similarly, $WTA_d = EG_c$ since the common decision is to give up the good in exchange for money. Thus we predict different values between points a and d, between a and c, and between b and d, and no significant difference between a and b and between c and d. The experimental results of Bateman et al. confirm statistically our predictions. Their econometric results indicate that, at least for the experiments where money was the numeraire, they cannot reject the equivalence of WTP_a and EL_b nor can they reject the equivalence of WTA_d and EG_c.[6]

4. Final remarks

In this paper, we presented an explanation of the divergence between WTP and WTA based on an agent's decision to purchase or sell a good under conditions of uncertainty, irreversibility, and learning over time. We discussed some pre-existing evidence from published experimental studies that provides intriguing empirical support for the existence of commitment costs in WTP and WTA. However, since these studies were not designed *ex ante* with specific hypotheses related to the commitment cost explanation in mind, further empirical verification is needed before it can be concluded that commitment costs are significant parts of observed WTP and WTA values.

[6] See Table 3 of Bateman et al. (1997) when money is used to elicit values. The authors also used a different response mode where Coke and chocolates are used as the unit of elicited values. They did observe different values for each reference point when the response mode is chocolate. However, errors of approximation may be significant since a subject could not choose a fraction of a chocolate bar. Moreover, their statistical test across the entire sample found no significant response mode effect (reported in their Table 5).

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